GPS LoS coverage

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***Abstract*—This report explains how to determine which GPS satellites are in line of sight (LoS) of a user located at EETAC (Castelldefels) at a given instant. The ECEF coordinates of every satellite are computed from the almanac published by CelesTrak; the position of the user is converted from geodetic to ECEF coordinates on the WGS-84 ellipsoid; the vector joining the user and each satellite is rotated to the local North-East-Down (NED) frame, and from it the distance, azimuth and elevation of the satellite are obtained. A satellite is considered usable if its elevation is at least 10°. For 12:00 UTC on 14/10/2026, nine satellites (PRN 5, 14, 15, 18, 20, 21, 22, 24 and 30) are visible. This is the same set that the online tool GNSS Mission Planning shows; the azimuths agree to within 0.4° and the elevations to within about 1° (3° for the satellite closest to the horizon). An interactive 24-hour sky plot has been added as an extension.**

***Index Terms*—GPS, line of sight, azimuth, elevation, NED, ECEF, sky plot.**

I. Introduction

When the user is at a specific location on the Earth's surface, it is important to know which subset of the GPS satellites is visible from that position at a given moment. A satellite is in LoS coverage if it is above the horizon of the user. To decide this, the pointing angles (azimuth and elevation) and the distance from the user to the satellite are needed [1].

This third session builds on the previous two. The functions written for Lab. Session 1 (`mean_anomaly`, `eccentric`, `true_anomaly`, `orbitcoords` and `satecef`) and the helper `almanac.m` of Lab. Session 2 give the ECEF coordinates of all the GPS satellites at any time, independently of the position of the user. Four new functions have been written: `kepler2ecef.m`, `lla2xyz.m`, `ecef2ned.m` and `ned2spherical.m`. They are called by the main program `main4.m`, which plots the visible satellites with the provided function `viewsat.m`. Section II describes the method step by step, Section III the interactive extension, and Section IV the results and their comparison with an online tool.

II. Method

Fig. 1 summarises the computation chain that `main4.m` executes for every satellite. The constants are the same as in the previous sessions: $G = 6.67384 \cdot 10^{-11}$ m^{3}kg^{-1}s^{-2}, $M_e = 5.972 \cdot 10^{24}$ kg, $\omega_e = 7.2921151467 \cdot 10^{-5}$ rad/s, and a tolerance of 10^{-8} for the Kepler equation.

A. Satellite coordinates in ECEF

The function `kepler2ecef.m` is `subsatellite.m` of the previous session without the last step: it returns the Cartesian ECEF coordinates (x, y, z) of the satellite at time "ToA $+ \Delta t$" instead of its latitude and longitude. For each row of the matrix `Eph` returned by `almanac.m` (eccentricity e, time of applicability t_{oa}, inclination i, node drift $\dot\Omega$, $a = (\sqrt{a})^2$, node longitude Ω_{week}, argument of perigee ω and mean anomaly M_0) the chain is

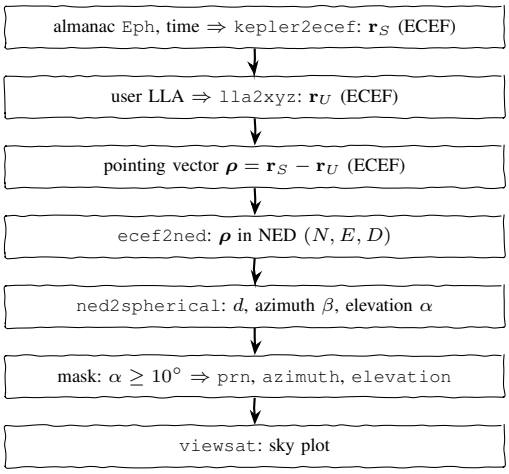


Fig. 1. Computation chain of `main4.m`.

$$\Delta t = t - t_{oa}, \qquad \Omega_0 = \Omega_{week} - \omega_e\, t_{oa}, \tag{1}$$

$$M = M_0 + \sqrt{GM_e/a^3}\;\Delta t, \tag{2}$$

$$E:\ E - e\sin E = M \quad \text{(Newton–Raphson)}, \tag{3}$$

$$\nu = 2\,\mathrm{atan2}\big(\sqrt{1+e}\sin\tfrac{E}{2},\ \sqrt{1-e}\cos\tfrac{E}{2}\big)\,, \tag{4}$$

$$x' = r\cos(\omega+\nu), \quad y' = r\sin(\omega+\nu), \quad r = a(1 - e\cos E), \tag{5}$$

$$\Omega = \Omega_0 + (\dot\Omega - \omega_e)\Delta t, \tag{6}$$

$$\begin{aligned} x &= x'\cos\Omega - y'\cos i\sin\Omega,\\ y &= x'\sin\Omega + y'\cos i\cos\Omega,\\ z &= y'\sin i. \end{aligned} \tag{7}$$

The time t is the number of seconds elapsed since the GPS week epoch (`esec`). The function `almanac.m` downloads the almanac that is valid at that time from CelesTrak [2] and returns `esec` for NOW, or for any UTC date and time if it is called with arguments ("frozen" time).

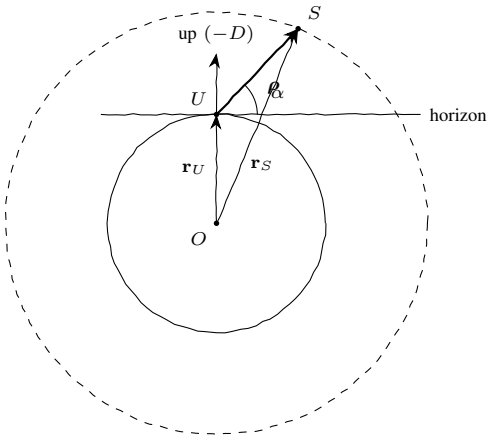


Fig. 2. Pointing vector $\boldsymbol{\rho}$ from the user U to the satellite S (side view) and elevation α above the local horizon. The solid circle is the Earth; the dashed one, the orbit.

B. Step 1: position of the user (`lla2xyz.m`)

The user is at EETAC. The geodetic coordinates read from the figure of the statement are latitude $\varphi = 41°16.578'$ N $= 41.27630°$, longitude $\lambda = 1°59.240'$ E $= 1.98733°$ and height $h = 51$ m. They are converted to ECEF on the WGS-84 ellipsoid ($a = 6378137$ m, $e^2 = 0.00669437999014$) with the radius of curvature in the prime vertical $N(\varphi)$:

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \tag{8}$$

$$\begin{aligned} x_U &= (N + h) \cos\varphi \cos\lambda, \\ y_U &= (N + h) \cos\varphi \sin\lambda, \\ z_U &= \big(N(1 - e^2) + h\big) \sin\varphi. \end{aligned} \tag{9}$$

The function takes the angles in radians and returns a 1×3 row vector. For EETAC the result is $\mathbf{r}_U = (4797555.076,\ 166472.410,\ 4185566.428)$ m.

C. Step 2: pointing vector

Since the user and the satellite are both expressed in the same Earth-fixed frame, the vector that points from the user to the satellite is simply the difference

$$\boldsymbol{\rho} = \mathbf{r}_S - \mathbf{r}_U, \tag{10}$$

as shown in Fig. 2. Its magnitude $d = \|\boldsymbol{\rho}\|$ is the geometric range. It can be computed for any satellite, also for those that are not in LoS, which will be useful in later sessions.

D. Step 3: ECEF to NED (`ecef2ned.m`)

The NED frame is centred at the user: N points to the geodetic north, E to the east and D downwards, along the local vertical. The rotation from ECEF to NED depends only on the latitude and longitude of the user:

$$\begin{bmatrix} N \\ E \\ D \end{bmatrix} = \underbrace{\begin{bmatrix} -\sin\varphi\cos\lambda & -\sin\varphi\sin\lambda & \cos\varphi \\ -\sin\lambda & \cos\lambda & 0 \\ -\cos\varphi\cos\lambda & -\cos\varphi\sin\lambda & -\sin\varphi \end{bmatrix}}_{R} \begin{bmatrix} \rho_x \\ \rho_y \\ \rho_z \end{bmatrix}. \tag{11}$$

The rows of R are the unit vectors north, east and down expressed in ECEF coordinates. The matrix that rotates from NED to ECEF, given in the NACC book [3], is its transpose, R^T, because a rotation matrix is orthogonal. As the program works with 1×3 row vectors (to simplify `fprintf`), the function transposes the input and the output: `ned = (R * ecef')'`.

E. Step 4: NED to spherical coordinates (`ned2spherical.m`)

From the NED components of $\boldsymbol{\rho}$, the distance, the elevation α and the azimuth β are

$$d = \sqrt{N^2 + E^2 + D^2}, \tag{12}$$

$$\alpha = \arcsin\left(-\frac{D}{d}\right), \tag{13}$$

$$\beta = \operatorname{atan2}(E,\, N) \quad (+360° \text{ if negative}). \tag{14}$$

The distance is equal to $\|\boldsymbol{\rho}\|$ because a rotation does not change the magnitude of a vector. The azimuth is measured clockwise from the north and goes from $0°$ to $360°$; `atan2` is used instead of `atan` to avoid the quadrant ambiguity. The elevation is negative if the satellite is below the horizon. Both angles are returned in degrees.

F. Step 5: visibility

A satellite is considered to provide a useful GPS signal if its elevation is at least $10°$ above the horizon. In `main4.m` three vectors are filled in the loop over the satellites: `prn`, `azimuth` and `elevation`, containing the identifier and the pointing angles of the satellites that fulfil this condition.

```
ecef_sat = kepler2ecef(Oe,G,Me,a,Mo,dt,e,tol,w,Oo,dO,io);
rho_ecef = ecef_sat - xyz_user;
ned      = ecef2ned(rho_ecef, phi, lambda);
[r, az, elev] = ned2spherical(ned);
if elev >= min_elev          % min_elev = 10
    prn = [prn, id];  azimuth = [azimuth, az];
    elevation = [elevation, elev];
end
```

G. Step 6: sky plot (`viewsat.m`)

The visible satellites are plotted on a polar plane centred at the user with the function `viewsat(azimuth, elevation, prn, min_elev, max_elev, downstr)`. The angle of the polar plot is the azimuth and the radius is proportional to $90° - \alpha$, so that the zenith is at the centre and the horizon on the outer circle. The pink ring is the region below the elevation mask and the dotted circles mark constant elevations.

H. Debugging file

`main4.dat` contains the debugging output of all the functions for every satellite at the instant of study: the parameters passed to `kepler2ecef` (`id`, Ω_0, `esec` and Δt), the intermediate variables of the orbit computation, the ECEF coordinates, the NED coordinates and the distance, azimuth and elevation. It ends with the list of visible satellites.

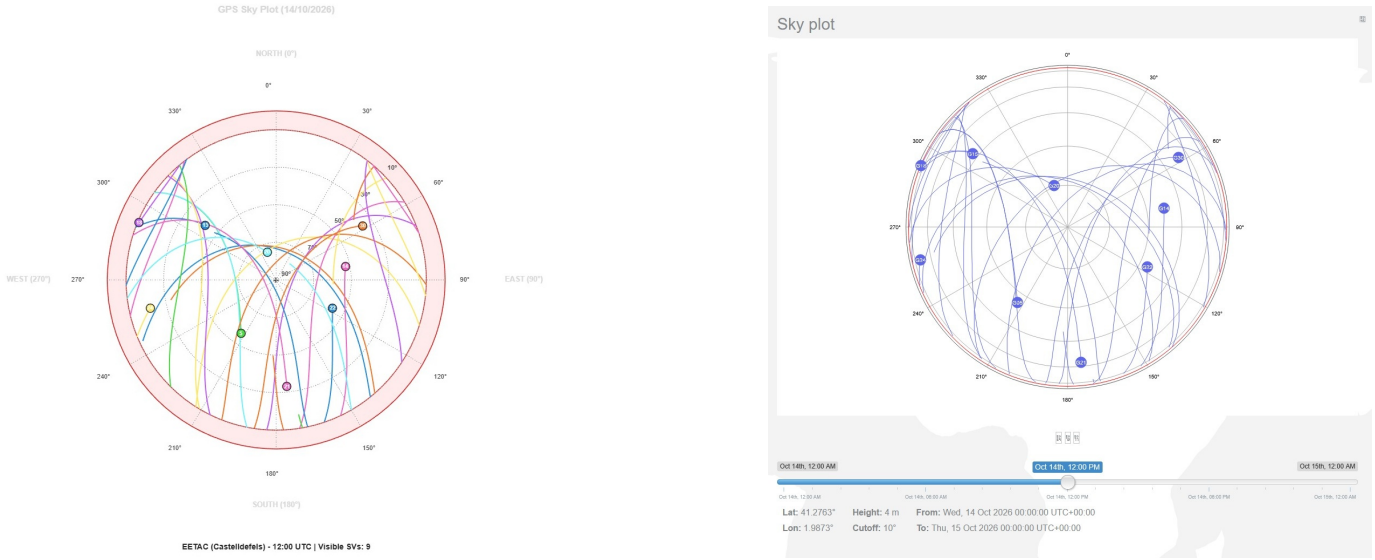


Fig. 3. Sky plot for EETAC at 12:00 UTC on 14/10/2026. Left: result of `main4.m` (the numbered circles are the visible satellites; the lines are their sky tracks). Right: capture of GNSS Mission Planning [4] (Castelldefels, cutoff 10°, GPS only, UTC time zone, 12:00 PM).

III. Extension: Interactive 24-Hour Sky Plot

Besides the steps required by the statement, `main4.m` offers an interactive sky plot. It precomputes the azimuth, elevation and range of all the satellites for the whole UTC day with a step of 1 minute (32×1441 evaluations of the chain of Fig. 1) and stores them in three matrices. A slider changes the time of the day; at each position the satellites above the $10°$ mask are drawn with their PRN, together with the part of their sky track that was above the mask since 00:00 UTC. Clicking on a satellite shows its azimuth, elevation and range. The time axis of the slider starts at 00:00 UTC of the chosen day, and the plot initially shows the instant of study.

IV. Results

A. Test case

The results are shown for 14/10/2026 at 12:00:00 UTC, the same instant that was used in the online tool. The program was run with the almanac of GPS week 392 (2440 including the two rollovers) and $t_{oa} = 319488$ s, with a frozen time ($e_{sec} = 302418.0$ s, $\Delta t = -17070$ s for all the satellites). The program can also be run for NOW, with `usar_ahora = true`. The almanac contains 32 satellites, 9 of which are above the $10°$ mask. Table I lists them with the values written to `main4.dat`.

TABLE I
Visible satellites at 12:00 UTC, 14/10/2026: computed values (`main4.dat`) and values read from GNSS Mission Planning [4].

	Computed			Online tool (read)	
PRN	Az. [°]	El. [°]	d [km]	Az. [°]	El. [°]
05	213.45	55.91	20989.7	213.8	55.6
14	78.97	52.48	21407.9	78.9	52.5
15	307.67	42.24	21408.2	307.5	41.7
18	292.68	10.84	24530.2	292.7	13.6
20	342.27	74.41	20420.5	342.5	73.5
21	174.24	32.94	22560.4	174.3	32.6
22	116.63	56.45	20998.5	117.0	56.5
24	257.39	21.27	23245.5	257.6	21.3
30	57.96	35.63	22457.1	57.9	35.3

TABLE II
Intermediate values written to `main4.dat` for PRN-05.

Quantity	Value
Ω_0 [rad]	1.891904
M, E, ν [rad]	1.232775, 1.237749, 1.242728
x', y' [km]	−24576.852, 9946.420
ECEF x, y, z [km]	24550.107, −5637.531, 8274.263
NED N, E, D [km]	−9817.154, −6485.503, −17381.857
d [km], β [°], α [°]	20989.693, 213.450, 55.905

B. Comparison with the online tool

Fig. 3 shows our sky plot next to the capture of GNSS Mission Planning [4], which was configured with the location of EETAC (latitude $41.2763°$, longitude $1.9873°$), a cutoff angle of $10°$, only the GPS constellation and the same instant. Both plots show exactly the same nine satellites (5, 14, 15, 18, 20, 21, 22, 24 and 30), and they are located in the same region of the sky: 20 near the zenith slightly to the north-west, 5 and 21 in the south, 14, 22 and 30 in the east, 15 and 24 in the west and 18 at the western edge, just above the mask.

To compare numbers, the azimuth and elevation of every satellite were measured on the capture by locating the centre of its marker. The azimuth is the angle around the centre of the plot and the elevation is obtained from the distance to the centre, which in that plot is proportional to $\cos\alpha$. The last two columns of Table I show that the azimuths agree to within $0.4°$ and the elevations to within about $1°$, which is the

precision of the measurement from a screen capture. The only larger difference is PRN 18 (10.8° computed, 13.6° read): it lies near the horizon, where one pixel of the plot corresponds to about one degree of elevation, and its marker overlaps with several tracks. In any case it is above the mask in both cases.

C. Discussion

The distance to the satellites ranges from 20420 km for PRN 20, which is almost at the zenith (74.4°), to 24530 km for PRN 18, which is close to the horizon. This is what is expected from the geometry: a satellite at the zenith is at a distance close to its altitude above the Earth (≈ 20200 km), and the distance increases as the elevation decreases, because the line of sight crosses a longer path. The other 23 satellites are not used: 3 of them are above the horizon but below the 10° mask, and 20 are below the horizon (negative elevation).

The height of the user (51 m in the program, 4 m in the online tool) has no practical effect on the pointing angles: changing it from 51 m to 4 m modifies the azimuth and the elevation of the satellites by less than 10^{-4} degrees.

V. Conclusions

- The LoS coverage of the GPS constellation from a given position has been computed with the chain ECEF position of the satellite, pointing vector, ECEF-to-NED rotation, and conversion to azimuth, elevation and distance. Only the latitude and longitude of the user are needed for the rotation.
- At 12:00 UTC on 14/10/2026, 9 of the 32 satellites are visible from EETAC with a 10° mask. Their sky positions agree with GNSS Mission Planning: the same satellites appear, with the azimuths to within 0.4° and the elevations to within the precision of the screen capture.
- The computation can be repeated for any time and date by changing the call to `almanac`, since the ECEF coordinates of the satellites do not depend on the user. The interactive sky plot uses this to show the whole day.
- The distance to each satellite is available as a by-product of the computation, for use in later sessions.

References

[1] J. Olmos, "Lab. Session 3: GPS LoS coverage," Lab statement, Navegació Aèria, Cartografia i Cosmografia, EETAC-UPC, 2025–26.

[2] CelesTrak, "GPS Yuma almanacs." [Online]. Available: https://celestrak.org/GPS/almanac/Yuma/

[3] D. Salazar, NACC course book, EETAC-UPC (available in Atenea), p. 59 (NED reference frame).

[4] GNSS Mission Planning, online satellite visibility calculator. [Online]. Available: http://gnssmissionplanning.com/